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INVERSELY DESIGNED PROJECTION IMAGE GENERATION FOR VAT PHOTOPOLYMERIZATION IN A MACHINE LEARNING FRAMEWORK

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ABSTRACT

In vat photopolymerization (VPP) processes such as mask-projection stereolithography (SLA) or digital light projection (DLP), oxygen dissolved within the photo-resin acts as an inhibitor and slows down photocuring kinetics. Since oxygen can diffuse from uncured areas to areas being irradiated, boundaries and fine features are slower to cure, resulting in an 'inhibitive' blurring effect. In contrast, incoming light can scatter when traveling through the photo-resin and locally cure undesired areas, resulting in both a 'facilitative' and inhibitive blurring effect. Both blurring effects lower the accuracy of the final print. In this work, we first gain an understanding of blurring effects by performing VPP printing of a single layer using a star shape. To model VPP, we then create a physics-based simulation discretized by pixel- and time-steps in Python to model VPP. To model both blurring effects, oxygen diffusivity and a Gaussian blur are applied to the simulation. An optimization algorithm is set up using PyTorch, an open-source machine learning framework, to optimize the projected shape to counteract the blurring effects and produce a more accurate print. Finally, we compare the fidelity of the optimized projection and the original non-optimized projection.

DMD	digital micromirror device
DoC	degree of conversion / degree of cure
FTIR	Fourier Transform Infrared
PSNR	Peak signal-to-noise ratio
Φ	concentration
SLA	stereolithography
SLM	spatial light modulator
σ	Gaussian blur standard deviation
VPP	vat photopolymerization

1. INTRODUCTION

Vat photopolymerization (VPP) is an additive manufacturing (AM) process in which a thin layer of liquid photocurable resin in a vat is selectively cured (solidified) by a light-activated polymerization process. After which, another layer of liquid resin is exposed and photo-cured, and layers are thus successively built in the z-direction to form the final part [1, 2]. VPP processes can differ in the manner by which light is directed onto the resin layer to cure the required shape, through: a) directing a small laser spot to move in the xy-plane to cure the layer pixel-by-pixel, or b) passing a uniform light through a digital micromirror device (DMD), spatial light modulator (SLM), or similar device to vary the light intensity at each given pixel, which is then projected onto the resin to cure the entire layer all at once [3, 4]. The former is called stereolithography (SLA), whereas the latter can be called digital light projection (DLP) or mask-projection. The working principles behind the first commercial VPP system were laid down in a seminal paper by Jacobs in 1992 [5], where they discovered a logarithmic relationship between depth of cure and exposure (with exposure, also called dosage, defined as intensity of incoming light

Keywords: Vat photopolymerization, machine learning, finite difference modeling, oxygen blurring effects, resolution optimization

NOMENCLATURE

AM	additive manufacturing
D	diffusivity
DLP	digital light projection

multiplied by time). This has since been called the Jacobs working curve [6] and has been shown to apply for several resins [7] although the reciprocity of intensity and time has been questioned for some other resins [8, 9].

The reaction mechanism of photopolymerization has been fully elucidated by Taki and Johnson in order to create models for the VPP process, consisting of initiation, propagation, and termination reactions [10-13]. The reaction kinetics are further complicated by the presence of dissolved oxygen in the resin, which reacts at least two orders of magnitude faster than the propagation reactions [14]. In effect, oxygen must be depleted before any curing can occur [15]. As such, oxygen is the primary cause of the induction time [16], or critical exposure as named by Jacobs [5], a threshold below which curing will not occur.

In a DLP system, oxygen can diffuse across shape boundaries, resulting in longer induction time and slower cure near the boundaries, visually observed as a reduced contrast and a blurring effect. Several strategies may be used to mitigate the blurring effects of oxygen [17]. Directly increasing the intensity of incoming light will minimize but not remove the induction time. Chemically altering the reaction mechanism by adding amines [18] can complicate kinetics. Degassing or evacuating the system prior to photopolymerization [19] can be time-consuming and expensive in practice on an industrial scale. If none of these strategies are adopted, oxygen will then have a spatial and time-dependent blurring effect on the resultant print as it diffuses across the boundaries of irradiation.

The projection of light into the resin vat has also been the subject of research since Jacobs [5], who also described their commercial SLA system's laser as a Gaussian distribution in the printing plane (xy-directions) with an exponential decay in the axial direction (z-direction) as per the Beer-Lambert law. In a projection-based system, each pixel will have a circle of influence in the printing plane which extends past the dimensions of the pixel, attributed to diffraction inefficiencies or otherwise, resulting in a Gaussian distribution. This phenomenon is also known as light bleeding or scattering [20, 21]. Like oxygen, light scattering has a blurring effect on the resultant prints. For the purposes of this paper, since scattering results in cure in areas without irradiation, this blurring effect is called a 'facilitative' effect; conversely, oxygen diffusion results in no cure in areas under irradiation, so this is called an 'inhibitive' effect.

Researchers have studied various ways to model this light scattering, with the most common being a point spread function described as a Gaussian distribution [21], a point spread function described with an experimentally determined distribution [22], or convolutions on an image [23]. Given a scattering model, optimization algorithms may be used to produce a projected image that accounts for the effects of scattering such that the resultant print will look closer to the target cure. For example, a bespoke algorithm was tested and resulted in lower error as quantified by custom formulas [22]. Linear programming and mixed integer programming models have been attempted, using an objective function that minimizes the error in the degree of cure (DoC) for each pixel [23] or minimizes the overall input energy while retaining cure in the target voxels [24], resulting in

improved cures. More recently, with the improvement of computational power, machine learning techniques have been applied to the optimization algorithm. Neural networks are most common with the drawback that large training data sets are required [25, 26], although data generation strategies may be used to expand the data set [27].

1.1 Basis and approach for this paper

The basis of this paper was prompted by an ongoing project in which we set up a VPP system using a resin containing 2.98wt% Irgacure 1173, 0.45wt% tetraphenylethylene, and 96.57wt% 2-phenoxyethyl acrylate; this resin was specifically chosen to be slow to cure. Despite using a uniformity correction mask as shown in Figure 1a, large blurring effects remain when comparing the projected images (Figure 1b, d) and the resultant cures (Figure 1c, e). We omit the details of the setup since that is not the focus of this paper.

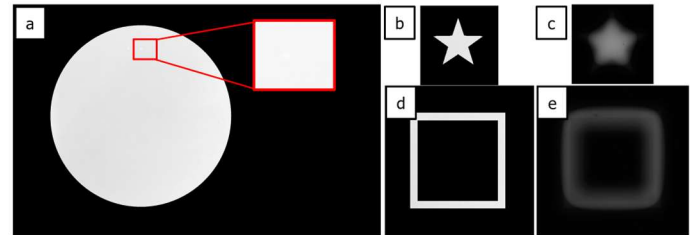


FIGURE 1: A) UNIFORMITY CORRECTION MASK SHOWING A CIRCLE IN WHICH ALL PROJECTED SHAPES SHOULD BE PLACED, SHOWING HIGH UNIFORMITY ACROSS THE AREAS USED IN THE DMD; INSERT IS A HIGHER-CONTRAST ZOOM. B) PROJECTED IMAGE OF A STAR SHAPE, WITH C) THE RESULTANT CURE BEING BLURRED. D) PROJECTED IMAGE OF A HOLLOW SQUARE SHAPE, WITH E) THE RESULTANT CURE BEING BLURRED. IMAGES ARE NOT TO SCALE.

In this paper, we set up a simulation of the blurring between the projected image and resultant cure, and a subsequent optimization model to counteract the blurring. We aim to achieve three goals: first, the simulation must use a chemistry-based reaction mechanism; second, the simulation must be able to run efficiently; and third, we must be able to directly apply backpropagation machine learning techniques to the simulation via PyTorch in order to optimize the projected image. Different levels of facilitative and inhibitive blurring are applied to visualize and quantify their effects on the cure. The fidelity of the resultant cure is compared to the original shape using peak signal-to-noise ratio (PSNR), a common image processing quality metric which represents pixel-by-pixel mean squared error in decibel form. The work is built on theoretical values and computational optimization alone, with experimental validation as a required future step.

2. MATERIALS AND METHODS

The simulation model is shown below in Figure 2 and will be described in the following sub-sections. All codes are written in Python, requiring PyTorch in the machine learning section.

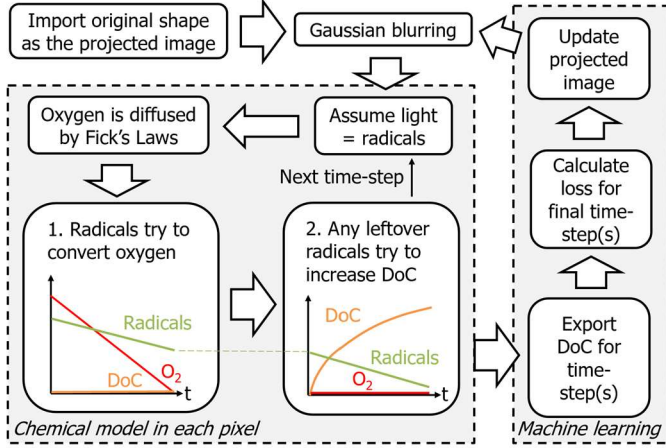


FIGURE 2: SIMULATION AND OPTIMIZATION MODEL FLOWCHART AS USED IN THIS PAPER. IN THE FIRST EPOCH, THE ORIGINAL TARGET SHAPE IS UPLOADED AS THE PROJECTED IMAGE. GAUSSIAN BLURRING IS CONDUCTED, FOLLOWED BY THE CHEMICAL MODEL THAT CALCULATES THE OXYGEN CONCENTRATION AND DEGREE OF CURE (DoC) FOR EACH TIME-STEP. ONCE THE CHEMICAL MODEL IS COMPLETE, THE MACHINE LEARNING MODEL IMPORTS THE DoC FOR THE CHOSEN TIME-STEP(S) AND CALCULATES LOSS AND UPDATES THE IMAGE FOR THE NEXT EPOCH.

2.1 Chemical model

We first note that the full elucidation by Taki [10] and Johnson [13] used upwards of 21 free parameters that needed to be determined experimentally, from literature, or optimized through other means such as trial-and-error, non-linear least squares method, particle swarm optimization, etc. Due to the complexity of such a model, we have opted to use a simplified chemical model in order to minimize the number of parameters which can be found via experiments in future work.

To start, light is assumed to convert linearly to photo-generated radicals. The radicals first convert oxygen molecules to peroxide in a linear fashion, requiring just one parameter value, K_1 , to describe the light-to-oxygen rate (gradient of slope), that can also vary linearly with intensity. The choice of a linear slope is arbitrary, and is used here solely to build up the model.

Second, any leftover radicals are then used to try and increase degree of conversion / cure (DoC). Real-time Fourier Transform Infrared (FTIR) spectroscopy can directly characterize the conversion of certain chemical groups over time [10], which is then used as a proxy for DoC. In our model, we use a normalized exponential function for our DoC, and we note the use of another parameter, K_2 , that describes the relative speed at which DoC increases over time for different intensity values (see equation 1).

$$\text{DoC} = 1 - \exp(-K_2 t^2) \text{ for } t > 0 \quad (1)$$

The DoC function is derived and adapted from [28, 29].

2.2 Molecular diffusion as an inhibitive effect

Oxygen diffusion is modeled based on Fick's Laws for 1D diffusion, which is well-described in modeling textbooks as below:

$$\phi_x^{t+1} = \frac{D\Delta t}{\Delta x^2} \phi_{x-1}^t + (1 - 2 \frac{D\Delta t}{\Delta x^2}) \phi_x^t + \frac{D\Delta t}{\Delta x^2} \phi_{x+1}^t \quad (2)$$

where Φ refers to the concentration of a particle or molecule in a medium, D is its diffusivity, and Δt and Δx are the discretized time and length interval steps. The superscript refers to the time; the subscript refers to the x -position. 1D solutions can be set up and solved via sparse n -diagonal matrices, for-loops, etc. in various programming languages.

This equation is easily extended into 2D and can be set up and solved in similar ways. However, in this case, we choose PyTorch and computer vision libraries for the problem setup due to their extensive open-source support from the machine learning community. The matrix of concentrations in 2D space are represented as a bitmap image, with 256 possible grayscale levels when using an 8-bit image. To avoid lossy compression, .bmp or .png files are used. Convolutions are then performed on the image using 3×3 kernels; this executes the same function as equation (2) but in 2D. The boundary conditions specified during the convolution will correspond to the various boundary conditions of the diffusion equation, i.e. a 'reflect' condition on convolution is equivalent to the Dirichlet boundary condition on the diffusion equation.

A well-understood phenomenon of numerical solutions to the diffusion equation is the Courant-Friedrichs-Lewy condition when multiple diffusion steps occur. For convergence and real-world interpretability of the results, the coefficient of ϕ_x^t , i.e. $(1 - 2 \frac{D\Delta t}{\Delta x^2})$ in the 1D case, must be less than or equal to 1 in an explicit solver. As such, timesteps must change in conjunction with the x -step; that is, if the pixel size gets smaller, there must be finer timesteps.

We note that oxygen is the smallest molecule in the system and thus has the greatest diffusivity. Consequently, only oxygen is specified to diffuse in our simulation; this improves runtime by ignoring the diffusion of larger molecules.

2.3 Light scattering as facilitative and inhibitive effects

Light scattering can be modeled efficiently by a Gaussian blur and requires only the standard deviation of the Gaussian distribution. In the image processing paradigm, this is yet another convolution with a kernel. The standard deviation (σ) is an integer multiple of the number of pixels across which the Gaussian blur is performed, although this can be generalized for arbitrary pixel sizes by multiplying σ with the pixel size for a generalized sigma σ_{gen} in length units. With the assumption that light scattering occurs the same way regardless of level of cure in the vat, we only need to perform this convolution once per simulation.

2.4 Machine learning model

The optimization is added as a wrapper around the light scattering and chemical model and is written in PyTorch [30], a

framework commonly used in machine learning. PyTorch is chosen for its ease of use, being open source, and compatibility with NVIDIA chips for faster runtimes. The model treats the pixels in the projected image as a rank 2 tensor, which is a state variable updated per epoch by gradient descent. We define a scalar loss value via a differentiable loss function, and PyTorch will then automatically determine the gradient for the loss function evaluated at the tensor and edit the pixel intensities for the next epoch to run. The Adam optimizer is used.

The loss function requires a careful definition of the target function. In broad terms, we highlight two requirements for optimizing the cure of a shape per Figure 3 below. First, we stipulate a relatively long cure window between timesteps t_1 and t_2 such that the operator has time to turn off the light. Second, we try to force the entire shape to cure together, that is, t_0 should be as close to t_1 as possible for a short developing cure time and long zero cure window.

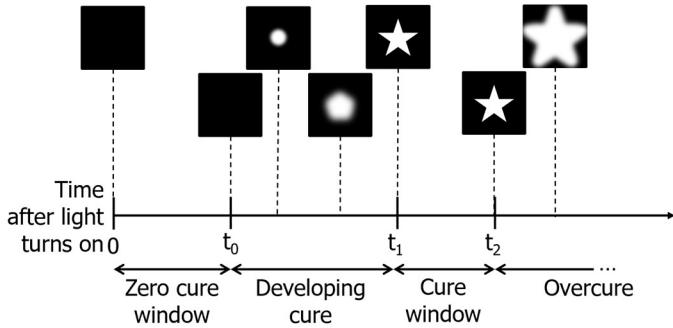


FIGURE 3: IDEAL TIMELINE OF OPTIMIZED CURE OF A STAR SHAPE.

The loss function is then the summation of errors at three timesteps: (1) the mean squared error between DoC and no cure at timestep t_0 , summed over all pixels, (2) the mean squared error between DoC and target shape (such as the star shape in Figure 3) at timestep t_1 , summed over all pixels, and (3) the mean squared error between DoC and target shape at timestep t_2 , summed over all pixels. The mean squared error can be efficiently computed with the Frobenius norm.

2.5 Summary of constants used in this model

Here we present a summary of all resin-specific and setup-specific parameters used in the chemical model. Other model-specific parameters such as timestep size or experiment-specific parameters such as light intensity are excluded.

First, K_1 in units of $(\text{mWcm}^{-2})^{-1}$ is an arbitrary constant used to vary the linear rate at which photo-generated radicals convert oxygen to peroxide as light intensity changes. As K_1 increases, the time at which cure begins will occur earlier.

Second, K_2 in units of s^{-2} is an arbitrary constant used to vary the exponential rate at which photo-generated radicals increase DoC as light intensity changes. A higher K_2 results in a shorter time between cure beginning and cure ending.

Third, D is the diffusivity of oxygen in the resin which controls the strength of the inhibitive blurring effect. A higher D

results in greater diffusion of oxygen over time, and hence a larger inhibitive blurring (i.e., cure does not occur despite incident light).

Fourth, σ_{gen} is the standard deviation of the Gaussian blur, but converted from a simulation-specific parameter without units to a non-simulation-specific length parameter. The larger the parameter, the greater the facilitative (i.e., cure occurs despite no incident light on the pixel) and inhibitive blurring.

These four parameters prove to be sufficient to achieve one of our goals in minimizing the number of parameters to be experimentally characterized.

3. RESULTS AND DISCUSSION

3.1 Success of optimization without considering facilitative or inhibitive effects

We initially set the model with both D and $\sigma_{\text{gen}} = 0$. The simulation worked to reduce intensity if the initial guess for target intensity was too high, but not the other way, as in Figure 4a. Since the target function relies on the DoC, we edit the DoC curve in each pixel (Figure 4b) by adding a small positive slope to the DoC in the zero cure window. This slope is a differentiable penalty term that encourages curing whenever the given timestep is too early for any curing to occur in the pixel, that is, whenever the intensity is too low for any curing to occur. Without this slope, the local loss topology would have a gradient of 0 and the optimizer would be unable to converge. (Due to the exponential function in equation 1, a similar slope exists near the $y=1$ asymptote.) We also note that a negative DoC has no physical meaning but is required for the machine learning model to recognize the need to increase intensity if it is too low. The edited DoC solves the issue.

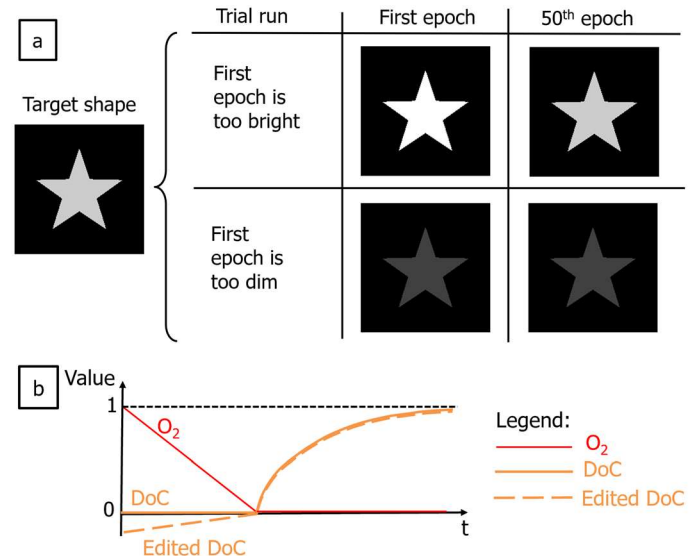


FIGURE 4: A) TWO TRIALS ARE CONDUCTED WHERE A DIFFERENT INITIAL GUESS IN THE FIRST EPOCH RESULTED IN THE OPTIMIZATION FAILING, B) EDITED O_2 AND DoC CURVE PER PIXEL, WHICH SOLVES THE PROBLEM.

3.2 Effects of adding oxygen diffusion

We add diffusivity to oxygen in order to observe its inhibitive effects without optimization (i.e., one epoch). Figure 5a shows the original target shape, with Figure 5b showing how the inhibitive effect decreased the overall size of the star and rounded and shaved off the corners of the star as highlighted in red. The peak signal-to-noise ratio (PSNR), which represents pixel-by-pixel mean squared error in decibel form, is 15.66. After optimizing 100 epochs, the optimized image in Figure 5c recognizes and corrects for the rounded corners by having a small amount of light immediately surrounding the corners as in Area A. The small amount of light photo-generates radicals to remove some oxygen in the area, reducing diffusion, but is dim enough to start curing only after the cure window. Additionally, edges of the star are sharpened as in Area B which perform the same function. The resultant DoC in the cure window is shown in Figure 5d, showing slightly better results visually and an improved PSNR of 22.91.

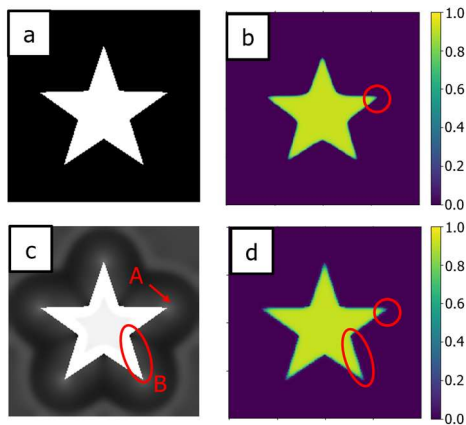


FIGURE 5: A) UNOPTIMIZED SHAPE, RESULTING IN B) COLORMAPPED DoC, C) OPTIMIZED SHAPE, RESULTING IN D) COLORMAPPED OPTIMIZED DoC.

We also note that one epoch takes approximately one to two seconds in a reasonably high-end laptop for an image of 200×200 pixels. The optimization procedure therefore takes about two minutes to complete, which is in line with one of our goals to run the software efficiently.

3.3 Effects of adding light scattering

We separately add the light scattering as a Gaussian blur to observe inhibitive and facilitative effects. Without optimization, as per the original image in Figure 6a and the projected blurred image in Figure 6b, the error in DoC as per Figure 6c is shown to be much higher than in the oxygen diffusion-only case. Even after optimization (original image in Figure 6d and projected blurred image in Figure 6e), the error in DoC as per Figure 6f is lower but still not as low as previously in the oxygen diffusion-only case. PSNR improves from 16.98 to 18.12 after optimization.

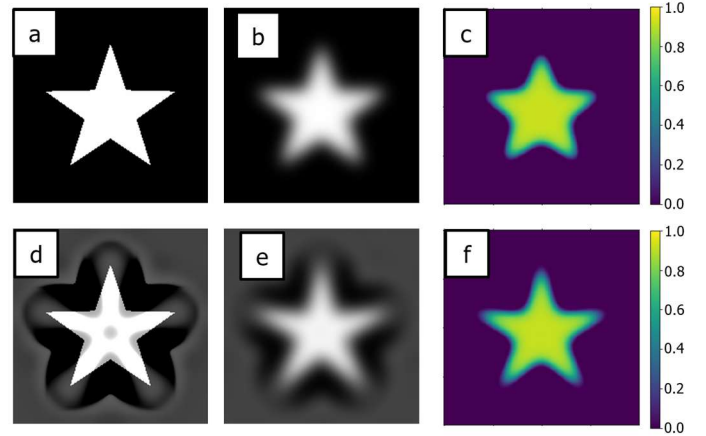


FIGURE 6: A) UNOPTIMIZED SHAPE, RESULTING IN B) PROJECTED SHAPE POST-GAUSSIAN BLUR AND C) COLORMAPPED DoC, D) OPTIMIZED SHAPE, RESULTING IN E) PROJECTED SHAPE POST-GAUSSIAN BLUR AND F) COLORMAPPED OPTIMIZED DoC. OXYGEN DIFFUSION DOES NOT OCCUR.

3.4 Effects of adding both oxygen diffusion and light scattering

Adding both oxygen diffusion and Gaussian blurring results in combined effects, as per Figure 7. The differences compared to Figure 6, that is, with light scattering but without oxygen diffusion, are hard to visually identify, suggesting that this particular value of σ_{gen} affected the resultant cure to a larger degree than D. PSNR improves from 15.16 to 17.82 after optimization.

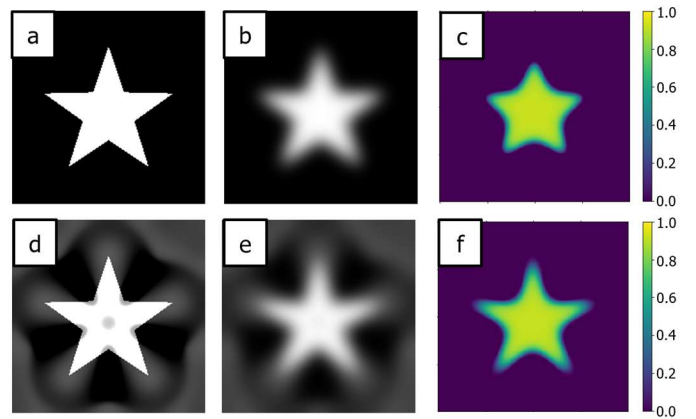


FIGURE 7: A) UNOPTIMIZED SHAPE, RESULTING IN B) PROJECTED SHAPE POST-GAUSSIAN BLUR AND C) COLORMAPPED DoC, D) OPTIMIZED SHAPE, RESULTING IN E) PROJECTED SHAPE POST-GAUSSIAN BLUR AND F) COLORMAPPED OPTIMIZED DoC. OXYGEN DIFFUSION OCCURS.

4. CONCLUSION

We successfully build and demonstrate an efficient simulation of one layer of VPP, accounting for the chemistry of the system and the oxygen diffusion that occurs. Using

theoretical values of light scattering and oxygen diffusivity, we find that inhibitive blurring effects can come from both oxygen diffusion and light scattering, and that facilitative blurring effects can come from light scattering. The simulation can be used to optimize the image, such that the resultant cure approximates the target image better with improved PSNRs. Future work includes determining the experimental levels of inhibitive and/or facilitative blurring and the determination of the four parameters used in this simulation, and expanding the simulation and optimization models into 3D VPP.

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